UNIT 6

UNITARY METHOD

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A process by which we find the value of a single unit from the value of multiple units and the value of multiple units from the value of a single unit.

Calculate the value of many objects of the same kind when the value of one of these objects is given.

Case 1
To find the value of many objects, we just multiply the value of one object with required number of objects.

Example 1
The price of a box is Rs 20. Find the price of 5 such boxes?
Solution:

The price of one box = 20 rupees
The price of 4 such boxes = (20 x 4) rupees = 80 rupees

Example 2
The price of one pen is Rs 3.50. What is the price of 6 such pencils?
Solution:

The price of 1 pen = 3.50 rupees
The price of 6 such pens = (3.50 x 6) rupees = 21 rupees

Case 2
To find the value of one object when the value of many objects is given, we divide the given value of the objects by the number of objects.

Example 3
The price of 4 kg mangoes is Rs 400. What is the price of 1 kg mangoes?
Solution:

The price of 4 kg mangoes = 400 rupees
The price of 1 kg mangoes = $\frac{400}{4}$ rupees = 100 rupees

EXERCISE 6.1

Q1. A notebook costs Rs 300. How much do 5 notebooks cost?
Solution:

Q2. A pencil costs Rs 50. What is the total cost for 12 pencils?
Solution:

Q3. A coffee mug is priced at Rs 800. What is the total price for 4 mugs?
Solution:

Q4. A toy car costs Rs 1,500. How much will 10 toy cars cost?
Solution:

Q5. A book costs Rs 1,000. Calculate the cost for 7 books.
Solution:

Q6. A single ticket to a concert costs Rs 3,000. What is the total for 5 tickets?
Solution:

Case 3:
Calculate the value of a number of same type of objects when the value of another of the same type is given (unitary method).

If value of many things is given. First we find the value of one thing and then find the value of the required number of things.

Example 1

Q. Amna purchased 6 books for Rs 180. How much will she pay for 14 such books?

Solution:

The price of 6 books $= \text{Rs } 180$

The price of 1 book $= \text{Rs } \frac{180}{6}$

The price of 14 books $= \text{Rs } \left(\frac{180}{6} \times 14\right)$

$= \text{Rs } (30 \times 14)$

$= \text{Rs } 420$

Hence, Amna will pay Rs 420 for purchasing 14 such books.

Example 2

Q. Shaina reads 80 pages of a novel in 4 hours. In how many hours can she read the book of 350 pages?

Solution:

For reading 80 pages, she takes $= 4$ hours

For reading of 1 page, she will take $= \frac{4 \text{ hours}}{80}$

For 350 pages, she takes $= \left(\frac{4 \times 350}{80}\right)$ hours

$= (20 \times 350)$ hours $= 17.5$ hour

So, Shaina can read the book in 17.5 hours.

EXERCISE 6.2

Q1. The price of 6 balls is 240 rupees. Find the price of 10 such balls?
Solution:

Q2. Rent of a house for 3 months is Rs 18,000. What is the rent for 8 months?
Solution:

Q3. The weight of 16 bags of price is 775.60 kg. What will be the weight of 24 such bags?
Solution:

Q4.The bus fare of 10 persons from Karachi to Larkana is 8,300. What is the fare for 36 persons?

Solution:

Q5.The cost of 10 books is Rs 240. What will be the cost of such 15 books?

Solution:

DIRECT AND INVERSE PROPORTION

Ratio:
A ratio is a way of comparing two or more quantities. Ratio of two numbers 8 and 2 is expressed as 8:2 or We read 8:2 as (8 is to 2).

For example: The ratio of ages of Ibrahim and Ismail is 2:1 then it shows that:
Ibrahim is older than Ismail
Ibrahim is twice as old as Ismail

Proportion:
An equation in which two ratios are set equal to each other. The symbol for proportions is "=" or ": :" If a: b and c: d are two ratios then the proportion between these two ratios is written as:

a: b = c: d or a:b :: c:d.

It is read as "Ratio a is to b is same as c is to d". Here a, b, c and d are respectively called as first term, second term, third term and four term of the proportion.

Example: 4 : 5 = 8 : 10 is a proportion.
We can write it as 4 : 5 :: 8 : 10

Types of Proportion:
There are two types of proportion:
1) Direct proportion
2) Inverse proportion

Direct Proportion:
The relation between two quantities where one value increases, so the other value also increases and vice versa, then the given quantities are said to be in direct proportion.

Examples:
More temperature and more heat; less temperature and less heat.
More energy and more work; less energy and less work.

Inverse Proportion:
The relation between two quantities where one value increases, and the other value decreases and vice versa, then the given quantities are said to be in inverse proportion.

Examples:
speed of the car increases, the time taken to cover certain distance decreases.
More buses on the road, less space on the road.

Identify and write direct or inverse proportion.

- More books, more money (Direct proportion)
- More apples, pay more money. (.........................)
- Less labourers, more time to build a house. (....................)
- Less workers, more time to do the work. (.............................)

Solve real life problems involving direct and inverse proportion (by unitary method)

Example 1
A man buys 4 kg of mangoes for Rs 360. How much will he pay for 9 kg of mangoes?
Solution:

Kg	Rs	Kg	Rs
4	360	9	$\boxed{}$

For 4 kg of mangoes, he spends Rs 360
For 9 kg of mangoes, he spends Rs $\frac{360}{4} \times 9$ = Rs 810

Example 2

6 labourers can build a house in 10 hours. How much time will 12 labourers take to build the same house?

Solution:

Labors	Hours	Labors	Hours
6	10	12	$\boxed{}$

6 labourers can build a house in = 10 hours

6 labourers can build a house in = 6 x 10 hours (Less labourers more time)

12 labourers can build the house in hours = $\frac{60 \text{ hours}}{12}$

(More labourers less time) = 5 hours

So, 12 labourers will build the same house in 5 hours.

EXERCISE 6.3

Q1. The ratio of pocket money of Ayesha and Aliza is 4:6. Write true or False.

(i) Ayesha has less pocket money than Aliza. (   )

(ii) Aliza has less pocket money than Ayesha. (   )

Q2. Express the following as ratio.

(i) 6 men to 9 women

Solution:

(ii) 4 litres to 10 litres

Solution:

(iii) 4 hours to 6 hours

Solution:

Q3. Identify and write direct or inverse proportion.

(i) More energy, more work. (.............)

(ii) More speed of a bike, less time to cover distance. (..............)

(iii) more money, more shopping. (.............)

(iv) More employees, more work done. (..............)

Q4. If 5 kg of apples cost Rs 100, how much will 8 kg of apples cost?

Solution:

Q5. A car travels 240 km on 12 litres of petrol. How far can it travel on 20 litres?

Solution:

Q6. A tank can be filled by a tap in 10 hours. How long will it take to fill the same tank with 3 taps working together if one tap can fill it in 10 hours?

Solution:

Q7. If 6 kg of rice costs Rs 120, how much will 15 kg of rice cost?

Solution: